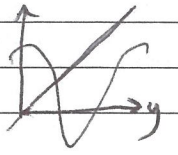
19/10/26

Plan

- Wednesday – tutorial, feedback sheet 2 → by noon on Friday
- Friday 23rd noon – midnight Summative Coursework 1 "Computer-based Test"



Q2 $y'' - 2y' - 3y = e^{2t}$

homog: $y'' - 2y' - 3y = 0$

Std ME $y = e^{Mt} \Rightarrow M^2 - 2M - 3 = 0$

$\Rightarrow (M-3)(M+1) = 0$

$\therefore y_c = Ae^{-t} + Be^{3t}$

Non-homog $y = Ce^{2t}$ since 2 is not a root of the characteristic eq:

$y' = 2Ce^{2t}$

$y'' = 4Ce^{2t}$

Substitute:

$4Ce^{2t} - 2(2Ce^{2t}) - 3Ce^{2t} = e^{2t}$

$-3C = 1 \quad \therefore C = -\frac{1}{3}$

$y = y_c + y_p$

$y_{g.s.} = Ae^{-t} + Be^{3t} - \frac{1}{3}e^{2t}$ //

$y(0) = 0 \Rightarrow 0 = A + B - \frac{1}{3}$

$y'(0) = 1 \Rightarrow -Ae^0 + 3Be^0 - \frac{2}{3}e^0 = 1$

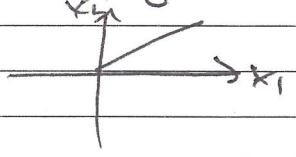
} solve for A and B.

Repeat evalues

- $\underline{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = e^{rt}\left(c_1 \underline{\xi}^{(1)} + c_2 \underline{\xi}^{(2)}\right)$

$$\frac{x_1}{x_2} = \frac{e^{rt}\left(c_1 \xi_1^{(1)} + c_2 \xi_1^{(2)}\right)}{e^{rt}\left(c_1 \xi_2^{(1)} + c_2 \xi_2^{(2)}\right)} = C$$

$x_2 = \frac{1}{C} x_1$



Matlab $x' = ax + by$, $y' = cx + dy$ $= \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix}$

$|A - r\underline{I}| = 0 \Rightarrow$ eigenvalues r_1, r_2

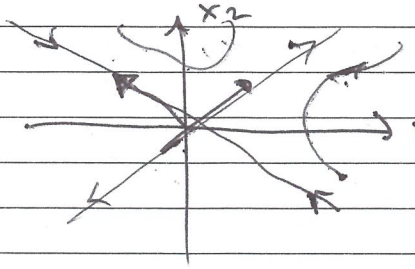
$A\underline{\xi} = r\underline{\xi} \rightarrow$ eigenvector $\underline{\xi}$

$t \to \infty$

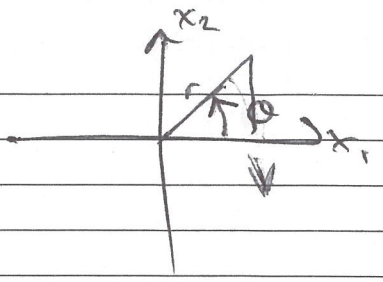
$t \to -\infty$ ...

$r_1 = r_2 = 2$

$\underline{\xi}^{(1)} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \longleftrightarrow \begin{pmatrix} 1 \\ -1 \end{pmatrix}$



Complex conjugate



$$r^2 = x_1^2 + x_2^2$$

$$2r\frac{dr}{dt} = 2x_1\frac{dx_1}{dt} + 2x_2\frac{dx_2}{dt}$$

$$\frac{dr}{dt} = \boxed{r' = \lambda r} \qquad \theta' = -\mu \quad (r \neq 0)$$

sep of var

$$\int \frac{dr}{r} = \int \lambda\, dt$$

$$\ln r = \lambda t + c_1$$

$$r = e^{\lambda t + c_1} = c_2 e^{\lambda t}$$

Next time – fundamental matrices